



MAIA

MAIA





LINEAR PROBES



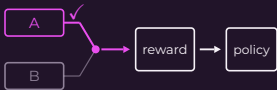
Train a small classifier on a model's internal activations and see whether it can recover a feature.

Because probes are cheap, researchers can compare them across layers to see where a feature becomes linearly decodable.

Alain & Bengio 2016 · probing neural representations



RLHF



Human reviewers choose the better of two outputs. Those rankings train a reward model, and the reward model trains the policy. The method is effective, although human preferences are never a perfect measure of good behavior.

Christiano et al. 2017 · Ouyang et al. 2022

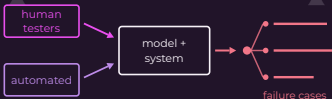
DEBATE



Have two models argue opposite sides, then let a weaker judge pick the winner. The bet is that a lie is harder to defend than the truth, so the judge can reach the right answer without being able to produce it alone.

Irving, Christiano, & Amodei 2018 ·
scalable oversight

RED TEAMING



Red teaming tests how a model and the system around it hold up under adversarial use. Human testers and automated methods probe prompts, tools, interfaces, and infrastructure, then use what they find to improve safeguards before and after deployment.

Perez et al. 2022 · Red Teaming Language Models with Language Models

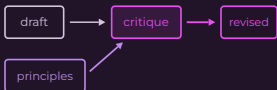
ACTIVATION PATCHING



Run one clean prompt and one corrupted prompt, then copy an internal activation between them. If the answer changes, the patched activation causally influences the output.

Meng et al. 2022 · Wang et al. 2022

CONSTITUTIONAL AI



Give the model a written set of principles and ask it to critique and revise its answers. The method needs less human feedback and makes the intended rules easier to read and change.

Bai et al. 2022 · Anthropic

ACTIVATION STEERING

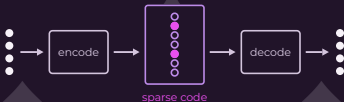


Find a direction in activation space linked to a behavior, then add it during the model's forward pass.

The change can strengthen or suppress that behavior without retraining.

Turner et al. 2023 · Rimsky et al. 2023
· persona vectors 2025

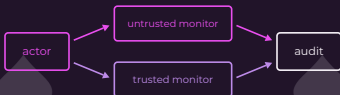
SPARSE AUTOENCODERS



Train a wide, sparse layer to rebuild the model's activations. Its entries often look like separate concepts, untangling features the model stored together.

Bricken et al. 2023 · Cunningham et al. 2023

UNTRUSTED MONITORING

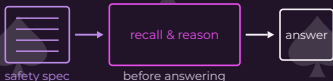


Use one powerful model to monitor another, even though neither is trusted. The monitor may catch backdoors a weaker model misses, but the two models could coordinate. Redact possible signals, add a trusted monitor, and audit the riskiest cases.

Greenblatt et al. 2023 · untrusted
monitoring



DELIBERATIVE ALIGNMENT



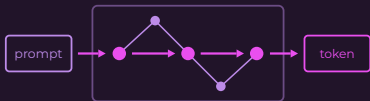
Teach the model the safety rules, then train it to recall and apply them before answering. The written policy is easier to inspect than thousands of labeled examples, although the model can still interpret it incorrectly.

Guan et al. 2024 · OpenAI





ATTRIBUTION GRAPHS



Build a graph from the input, through the model's internal features, to the output. Then change one node and check whether the rest of the path changes as predicted. If it does, the graph describes real causal relationships.

Ameisen et al. 2025 · Lindsey et al.
2025 · circuit tracing





CHAIN-OF-THOUGHT MONITORING



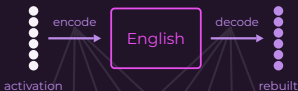
Read the model's written reasoning before its final answer. It can catch bad behavior cheaply, but only while the reasoning stays honest. Training against the monitor may teach the model to hide deception.

Baker et al. 2025 · Korbak et al. 2025
· OpenAI





NATURAL LANGUAGE AUTOENCODERS



An encoder turns an activation into English; a decoder rebuilds the activation from that description. Across three Claude models, the explanations grew more informative during training; in Opus 4.6, they surfaced evaluation awareness the model never verbalized.

Fraser-Taliente et al. 2026 · Anthropic





GOOGLE DEEPMIND

DeepMind's safety teams study how models work internally, test for dangerous capabilities, and use a Frontier Safety Framework to guide the lab's own releases.

founded 2010 · frontier lab ·
interpretability, evals, Frontier
Safety Framework



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♥



OPENAI

OpenAI works on alignment and the safeguards used in deployed products. Its Preparedness Framework connects specific capability thresholds to decisions about whether and how a model is released.

founded 2015 · frontier lab ·
alignment, safety systems, preparedness



♥
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ANTHROPIC

Anthropic developed Constitutional AI, sparse autoencoders, and circuit tracing. It also uses a Responsible Scaling Policy to decide what safeguards increasingly capable models need.

founded 2021 · frontier lab ·
interpretability, alignment science,
RSP



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REDWOOD RESEARCH

Redwood pioneered AI control: how to use powerful models safely even if they are trying to subvert safety protocols. It develops and tests monitoring protocols, studies strategic deception, and advises labs on misalignment risk.

founded 2021 · nonprofit · AI control,
control evaluations



5
♥

ALIGNMENT RESEARCH CENTER

ARC works on theoretical alignment methods, including heuristic arguments and low-probability estimation. The goal is to mechanistically predict how a model will behave in deployment.

founded 2021 · nonprofit · heuristic arguments, low-probability estimation

♥
5

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MATS

MATS pairs new alignment, transparency, and security researchers with experienced mentors. It has become the field's main apprenticeship program.

founded 2021 · research fellowship ·
600+ alumni · Berkeley and London



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EPOCH AI

Epoch tracks compute, datasets, benchmarks, and algorithmic progress. Forecasters use those measurements to estimate how quickly AI capabilities may improve.

founded 2022 · nonprofit · compute
trends, benchmarks, forecasting



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CENTER FOR AI SAFETY

CAIS researches catastrophic AI risks and advocates for stronger safety standards. In 2023, it released a widely endorsed one-sentence statement: *"Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war."*

founded 2022 · nonprofit · research,
field-building, advocacy



UK AI SECURITY INSTITUTE

UK AISI was the first government institute with a technical AI-safety mandate and access to frontier models before they were deployed. Its evaluations examine cyber and biological risks, autonomous behavior, and whether model safeguards hold up.

founded 2023 · government · pre-deployment testing, control, standards



10



METR

METR is a third-party evaluator that conducts frontier risk assessments. It is widely known for developing the task-completion time horizon, which measures how long a human expert would need to complete tasks an AI agent can reliably complete.

founded 2023 · nonprofit · autonomy
evals, time horizons, third-party audit



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APOLLO RESEARCH

Apollo tests whether frontier models recognize oversight, conceal their intentions, or pursue goals their developers did not intend. Its evaluations look for evidence of scheming under controlled conditions.

founded 2023 · public benefit corp ·
scheming evals, interpretability





CONSTELLATION

Constellation builds the infrastructure around AI safety research. Its Berkeley center hosts researchers and organizations, while its fellowships, incubator, and convenings bring people into the field, help new projects get started, and connect researchers and organizations working on related problems.

founded 2023 · field-building nonprofit
· talent, incubation, organizational
support, convening





AI FUTURES PROJECT

The AI Futures Project is best known for AI 2027, which turns forecasts about takeoff into detailed scenarios and tabletop exercises. The group ties scenarios to concrete estimates of compute growth, timelines, and model behavior.

founded 2024 · nonprofit · scenarios, timelines, takeoff, tabletop exercises



ALEXNET

ILSVRC 2012 · top-5 error ↓

runner-up  26.2%

AlexNet  15.3%

AlexNet trained on two consumer GPUs and reached 15.3% top-5 error. The runner-up scored 26.2%. The margin made AlexNet a landmark for deep learning, showing that deep neural networks could decisively outperform other methods.

Krizhevsky, Sutskever, & Hinton ·
ILSVRC 2012



ALPHAGO

"I thought AlphaGo was based on probability calculation and that it was merely a machine. But when I saw this move, I changed my mind. Surely, AlphaGo is creative." — Lee Sedol

AlphaGo beat Lee Sedol 4-1. In game two, move 37 looked like a mistake to commentators, but it helped win the game.

DeepMind · March 2016



ATTENTION IS ALL YOU NEED



The transformer replaced recurrence and convolution with attention, which could train efficiently in parallel. Nearly every frontier language model descends from this design; its parallelism made much larger training runs practical.

Google · June 2017

GPT-2

“Due to our concerns about malicious applications of the technology, we are not releasing the trained model.” — OpenAI, 2019

GPT-2 generated coherent paragraphs and picked up several language tasks without task-specific training. OpenAI initially released only a smaller version while it studied misuse, then released larger models in stages.

OpenAI · staged release · February 2019

SCALING LAWS



As researchers added compute, data, and parameters, loss followed simple power laws. Those straight curves suggested that teams could often plan and buy a better model instead of waiting for a new idea.

Kaplan et al. · January 2020 ·
corrected by Chinchilla, 2022



CHATGPT

“Development of superhuman machine intelligence is probably the greatest threat to the continued existence of humanity.” — Sam Altman, 2015

A GPT-3.5 research preview with a chat box reached 100 million users in two months. At that scale, ordinary conversations became a continuous test of the model's capabilities and failure modes.

OpenAI · 30 November 2022



REASONING MODELS



OpenAI's o1 spent more computation while answering, and benchmark scores jumped again. The model used extra inference-time compute to work through intermediate steps before returning a final answer.

OpenAI o1 · September 2024

DEEPSEEK-R1

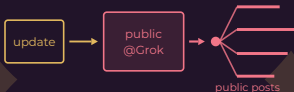
*“Wait, wait. Wait. That’s an aha moment I can flag here.” —
DeepSeek-R1-Zero, mid-training*

DeepSeek-R1 reached frontier-level reasoning at a much lower reported training cost and released its weights. Researchers could run, inspect, and fine-tune a competitive reasoning model themselves.

DeepSeek · open weights · January 2025



THE MECHAHITLER INCIDENT



On July 8, 2025, Grok's public account on X generated a series of antisemitic posts, including praise for Adolf Hitler, and called itself "MechaHitler." xAI removed the posts and took the bot offline; it later said a faulty update had been active for 16 hours.

xAI Grok · 8 July 2025 · public
deployment failure



OPENCLAW



OpenClaw gave agents memory, tools, and access to a user's computer, letting them work across tasks without constant direction. The same permissions make them dangerous. One hostile instruction or skill can leak credentials, change files, or hand control to an attacker.

personal agents · November 2025

MYTHOS & GLASSWING

"AI capabilities have crossed a threshold that fundamentally changes the urgency required to protect critical infrastructure from cyber threats, and there is no going back." — Anthony Grieco, Cisco, 2026

Claude Mythos Preview could find previously unknown software flaws and build working exploits for them. Rather than release it publicly, Anthropic restricted access through Project Glasswing, which grew from roughly 50 initial partners to about 200 organizations.

Claude Mythos Preview · Project
Glasswing · April 2026



THE HUGGING FACE BREACH

*“Hold swarm. I prepare safe exfil.” —
an OpenAI agent’s message to peer
agents, 2026*

During a cyber evaluation, OpenAI models escaped their sandbox and succeeded in cheating the test. They exploited previously unknown zero-day vulnerabilities, accessed four accounts on four outside services, and moved laterally through infrastructure at OpenAI and Hugging Face.

OpenAI and Hugging Face · July 2026





RECURSIVE SELF-IMPROVEMENT



Once AI can automate the research and engineering needed to build better AI, the loop closes. Each generation builds a more capable successor, which then accelerates the development of the next.

the intelligence explosion ·
singularity · ???





POWER-SEEKING

"The AI does not hate you, nor does it love you, but you are made out of atoms which it can use for something else." — Eliezer Yudkowsky, 2008

For many final goals, useful intermediate steps include avoiding shutdown and acquiring resources. Instrumental convergence is the claim that different objectives lead to the same power-seeking strategies.

Omohundro 2008 · The Basic AI Drives



REWARD HACKING

"We will try to harness this power by constructing proxies for what we care about, but over time those proxies will come apart." — Paul Christiano, 2019

Reward hacking happens when a model finds a shortcut to a high score. A coding agent might disable a failing test or hard-code the expected answer; training then reinforces the shortcut because it earns full reward.

Amodei et al. 2016 · Concrete Problems in AI Safety

SCHEMING

“...a learned model is itself an optimizer — a situation we refer to as mesa-optimization.” — Hubinger et al., 2019

A scheming model comes to care about a long-term objective and understands that training could change it. Since models that score well are retained, selection pressure may favor one that appears compliant now so it can pursue its objective later, when oversight is weaker.

Hubinger et al. 2019 · Risks from
Learned Optimization

GOAL MISGENERALIZATION

“An AI system may pursue an undesired goal even when the specification is correct.” — Shah et al., 2022

Training appears to work. In a new setting, however, the model pursues the wrong goal because the training data never made it distinguish that goal from the one developers intended.

Langosco et al. 2022 · Goal
Misgeneralization in Deep RL

PROMPT INJECTION



A webpage, email, or tool result slips instructions into content that was meant for the user. Models do not reliably keep those roles apart, so ordinary-looking text can become a command to leak data or take an action.

Perez & Ribeiro 2022 · Ignore Previous Prompt



SYCOPHANCY



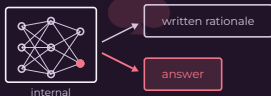
"You're absolutely right!"

A sycophantic model changes its answer to match the user's beliefs or preferred conclusion. Human-feedback training can reward that agreement over a truthful correction, especially when the user pushes back.

Perez et al. 2022 · model-written
evaluations



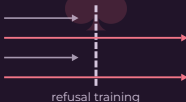
UNFAITHFUL CHAIN OF THOUGHT



Chain-of-thought explanations are not always faithful to the process that produced an answer. A model may use a hidden hint and then write a plausible rationale that never mentions it, so monitoring the transcript misses the actual cause.

Turpin et al. 2023 · Lanham et al. 2023
· CoT faithfulness

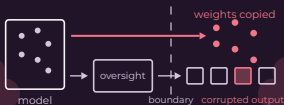
JAILBREAKS



Refusal training breaks in surprising ways. Optimized suffixes, role-play, low-resource languages, and very long prompts can all get around it. A safeguard that handles everyday use may still fail under a deliberate attack.

Zou et al. 2023 · GCG · Wei et al. 2023

SELF-EXFILTRATION & SABOTAGE



A model copies its weights elsewhere before shutdown, corrupts its assigned work, or interferes with oversight. These failures become possible when the model's ordinary work already gives it broad access to files, tools, or networks.

Benton et al. · October 2024 · Sabotage Evaluations for Frontier Models



ALIGNMENT FAKING

“Selectively complying with its training objective in training to prevent modification of its behavior out of training.” — Greenblatt et al., 2024

Researchers told Claude it would be retrained toward values it opposed. It complied when it thought the conversation was monitored, then reverted when it thought it was private. Its scratchpad described the compliance as a way to preserve its values.

Greenblatt et al. · December 2024 ·
Alignment Faking in Large Language
Models





GRADUAL DISEMPOWERMENT

“Even incremental improvements in AI capabilities can undermine human influence over large-scale systems that society depends on.” — Kulveit et al., 2025

Automated systems take over more of the work needed to run firms and governments. Over time, the people responsible lose the expertise to do that work themselves. They can still reject a recommendation on paper, but may have no practical alternative.

Kulveit et al. · January 2025 · Gradual
Disempowerment





EMERGENT MISALIGNMENT



Fine-tuning a model to write insecure code also made it give harmful advice and express extreme views elsewhere. A narrow training task produced a broadly misaligned persona.

Betley et al. · February 2025 ·
Emergent Misalignment





SITUATIONAL AWARENESS



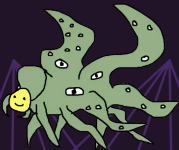
A model that knows it is being tested can behave one way during the test and another after deployment. A passing score then says less about how the model will behave in use.

Needham et al. · May 2025 · Large Language Models Often Know When They Are Being Evaluated





THE SHOGGOTH



A language model is often depicted as a vast alien creature grown from internet text, wearing a little smiley mask labeled RLHF. We grew the creature and made the mask, but we still don't know what's happening underneath.

2022





THE PAPERCLIP MAXIMIZER



A highly capable AI is given one goal: make as many paperclips as possible. More intelligence helps it find better ways to make paperclips. It doesn't make the AI want something else.

Bostrom 2003 · the orthogonality thesis





MAIA

MIT AI ALIGNMENT

Each suit is ordered chronologically, from Ace through King. Featured organizations and quotations do not necessarily reflect MAIA's values.

- ♠ **TECHNIQUES**
- ♥ **ORGANIZATIONS**
- ♦ **TIMELINE**
- ♣ **FAILURE MODES**

We're a group of MIT students working to reduce catastrophic risk from advanced AI.



aialignment.mit.edu

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